

[PHYS 6268] Nonlinear Dynamics: Problem Set 11

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Strogatz Book: Problems 9.3.1, 9.3.4, 9.3.8, & 9.3.9

1 Problem 9.3.1: Quasiperiodicity $\neq$ Chaos

The trajectories of the quasiperiodic system

$$\dot{\theta}_1 = \omega_1 \quad ; \quad \dot{\theta}_2 = \omega_2 \quad ,$$

for $(\omega_1/\omega_2) \notin \mathbb{Q}$ are not periodic.

1.1 Part (a): Why isn't this system considered chaotic?

The system $(\dot{\theta}_1, \dot{\theta}_2)$ is most intuitively represented in a phase space configured as a torus since it possesses two unique, periodic coordinates. For an irrational ratio of system frequencies ω_1/ω_2 , the evolution of trajectories in this system will continue indefinitely around the torus repeatedly. Such trajectories will never self-intersect or repeat periodically, since there is never a least-common multiple of the periods for which the trajectory may return to its previous location. Therefore, the trajectories for this system will continue around the torus, winding through phase space as $t \rightarrow \infty$. Eventually, this ergodic system will “cover” the entire torus. Thus, the set of points in phase space defined by the trajectory is dense on the 2-torus. This implies that every trajectory, after some time t , will have passed arbitrarily close to each point on the torus.

Though the system never passes through the same coordinates on the phase space torus, its motion is predictable (for initial conditions (ϕ_1, ϕ_2)):

$$\begin{aligned}\theta_1(t) &= \omega_1 t + \phi_1 \pmod{2\pi}, \\ \theta_2(t) &= \omega_2 t + \phi_2 \pmod{2\pi}.\end{aligned}$$

Here the modulo operator $\pmod$ restricts the values of θ to the torus interval $[0, 2\pi]$. Not only is the system readily integrated for the indefinite behavior, but its evolution is quasiperiodic. Quasiperiodicity is a long-term behavior that is unique to the phase space torus, characterized by trajectories which never self-intersect, but also never close. Strogatz defines chaos by stating that

“Chaos is aperiodic long-term behavior in a deterministic system that exhibits sensitive dependence on initial conditions.”

Strogatz furthermore emphasizes that for a system to be chaotic, its trajectories must *never* “settle down”: since the system described by Problem 1 is quasiperiodic, trajectories are always “settled down” to the repeated motion that characterizes quasiperiodicity. Hence, even though trajectories are not periodic, they are still *repetitive*, consistently winding around the torus at a constant (but irrational) ratio of system frequencies ω_1/ω_2 . This caveat is why quasiperiodic systems such as these are *not* considered chaotic dynamical systems.

1.2 Part (b): Without using a computer, find the largest Lyapunov exponent for the system

n Liapunov exponents for a dynamical system of n dimensions are defined to be the values λ_k ($k \in [1, \dots, n]$) which describe the exponentially-growing deviation along the k direction of a sphere of trajectories that are infinitesimally separated by $|\delta_0| = |\delta(t=0)|$ at some initial time $t=0$. Explicitly, we let the difference in phase space coordinates along the k -th dimension be represented

by the time-dependent quantity $\delta_k(t)$. Then the separation evolves as

$$\delta_k(t) \sim \delta_k(0)e^{\lambda_k t} \quad ,$$

where λ_k is the Lyapunov exponent for the k -th principal axis of the deformed n -sphere. As t grows, the spheroid of trajectories is deformed into an ellipsoid, and the dominant Lyapunov exponent will eventually cause the ellipsoid to become flat. The diameter of this resultant quasi-ellipse is controlled by the *largest* Lyapunov exponent.

Hence, to calculate the largest Lyapunov exponent of the system, we evaluate

$$\lambda_{\max} = \lim_{t \rightarrow \infty} \lim_{\delta_0 \rightarrow 0} \frac{1}{t} \ln \left(\frac{|\delta(t)|}{|\delta_0|} \right) \quad . \quad (1)$$

We let $\omega = (\omega_1, \omega_2)$, so that

$$\theta(t) = \omega t + \phi \quad .$$

Our initial conditions are given by ϕ^i and ϕ^j for the two nearby trajectories respectively. These quantities satisfy $|\phi^i - \phi^j| < \epsilon$ for some arbitrarily small parameter $\epsilon \in \mathbb{R}$. However, as the system evolves, their positions separate. This separation can be calculated by writing the expression for each (vector) trajectory: for instance,

$$\theta^i(t) = \omega t + \phi^i \mod 2\pi \quad .$$

Then, in order to evaluate the limit for $\lambda_{\max}$, we calculate

$$\begin{aligned} |\theta^i(t) - \theta^j(t)| &= |(\omega t + \phi^i) \mod 2\pi - (\omega t + \phi^j) \mod 2\pi| \\ \text{let } C_k &\equiv ((\omega t + \phi^k) \mod 2\pi) \quad k = i, j \\ &\implies |\theta^i(t) - \theta^j(t)| = |C_i - C_j| \\ &\quad \text{let } m_i, m_j \in \mathbb{Z}^2. \\ &\implies C_k + 2\pi m_k = \omega t + \phi^k \\ &\implies (\omega t + \phi^i) - (\omega t + \phi^j) = (2\pi m_i + C_i) - (2\pi m_j + C_j) \\ &\implies \phi^i - \phi^j = 2\pi(m_i - m_j) + (C_i - C_j) \\ &\implies C_i - C_j = (\phi^i - \phi^j) - 2\pi(m_i - m_j) \\ &\implies |\theta^i(t) - \theta^j(t)| = |(\phi^i - \phi^j) - 2\pi(m_i - m_j)| \end{aligned}$$

Here the integers m_i, m_j are 2-vectors in order to be compatible with the arithmetic operations on the other 2 dimensional vectors. Hence, after manipulating the modulo representation of the trajectory, we have arrived at an expression that we can use to calculate $\lambda_{\max}$. Substituting into equation (1), we find

$$\lambda_{\max} = \lim_{t \rightarrow \infty} \lim_{(\phi^i - \phi^j) \rightarrow 0} \frac{1}{t} \ln \left(\frac{|(\phi^i - \phi^j) - 2\pi(m_i - m_j)|}{|(\phi^i - \phi^j)|} \right) \quad . \quad (2)$$

Then, if we let $(\phi^i - \phi^j) \rightarrow \delta$, we can write the Lyapunov exponent expression in a digestible form:

$$\lambda_{\max} = \lim_{t \rightarrow \infty} \lim_{\delta \rightarrow 0} \frac{1}{t} \ln \left(\frac{|\delta - 2\pi(m_i - m_j)|}{|\delta|} \right) \quad . \quad (3)$$

Hence, by evaluating the indeterminate limit above, we find that the maximal Lyapunov exponent is either $\lambda_{\max} = \ln |(2\pi(m_i - m_j))|$ for some integers m_i, m_j , or it vanishes with the limit. We find that depending on the order of evaluation for the limits (i.e., depending on the direction of approach in (x, t) space), either one of these two results is possible. Therefore, the minimum nonzero value of this expression is $\lambda_{\max} = \ln 2\pi$. However, in general, the maximal Lyapunov exponent is given by the limit above.

2 Problem 9.3.4: Numerical Experiments

For each of the values of r given below, use a computer to explore the dynamics of the Lorenz system, assuming $\sigma = 10$ and $b = 8/3$ as usual. In each case, plot $x(t)$, $y(t)$ and $z(t)$. Investigate the consequences of choosing different initial conditions and lengths of integration. Also, in some cases you may want to ignore the transient behavior and plot only the sustained long-term behavior.

r = 24.5:

In order to investigate these dynamics, we compute trajectories $\mathbf{x}(t) = (x(t), y(t), z(t))$ according to the Lorenz system:

$$\dot{x} = \sigma(y - x) \quad (4)$$

$$\dot{y} = rx - y - xz \quad (5)$$

$$\dot{z} = xy - bz \quad (6)$$

$$(7)$$

To do so, for an arbitrary coordinate of initial conditions $\mathbf{x}_0 = (x(t=0), y(t=0), z(t=0))$ we compute the vector of time derivatives $\dot{\mathbf{x}} = (\dot{x}, \dot{y}, \dot{z})$. In order to advance the system forward by an increment of time Δt , we make an explicit, forward-integrated (i.e., Euler-Forward) step assuming sufficiently small Δt to preserve numerical stability. This takes the form of

$$\mathbf{x}(\Delta t) = \mathbf{x}(0) + \dot{\mathbf{x}}(0) \Delta t \quad ,$$

which can be readily extended to a general t :

$$\mathbf{x}(t + \Delta t) = \mathbf{x}(t) + \dot{\mathbf{x}}(t) \Delta t \quad .$$

By iteratively computing this expression for a sufficiently small $\Delta t \ll 1$, this method provides a means to compute trajectories through phase space for the Lorenz system numerically across an arbitrarily large time interval. This routine is implemented as an explicit, forward-time scheme, so its stability criterion is sensitive to the choice of timestep Δt . Hence, we employ an adaptive algorithm that uses a higher order Runge-Kutta algorithm when the error for the explicit, first order forward integration exceeds a specified threshold. We employ a value of $\Delta t = 10^{-4}$ in nondimensional time units for all Lorenz system trajectory integrations where it is not otherwise specified. Therefore, our approach compromises between computational efficiency and sufficiently accurate solutions to the ODE system. This summarizes the computational procedure that is implemented to calculate Lorenz trajectories and visualize them in this section and in section 4.

To begin, we compute trajectories for the system using the “typical” slate of parameters for the Lorenz attractor: $r = 28$, $\sigma = 10$, and $b = 8/3$. The results of this trajectory computation (using the above integration procedure) are shown in Figure 1. As discussed in Lecture and in Strogatz, the Lorenz system admits three fixed points when $r > 1$. Two are (unstable) saddle points that lie symmetric across x and with positive z values corresponding to $z = r - 1$, in addition to an unstable saddle point at the origin. Precisely, these two saddle point equilibria are located at at

$$\begin{aligned} x^* &= \pm \sqrt{b(r-1)} \\ y^* &= \pm \sqrt{b(r-1)} \\ z^* &= r - 1 \quad . \end{aligned}$$

Analogous to the expected behavior, the system displays seemingly erratic “switching” between cycling around two separate equilibria in both $x(t)$ (Figure 1(a)) and $y(t)$ (Figure 1(b)). This behavior corresponds to the orbits looping around the two fixed points away from the origin, as

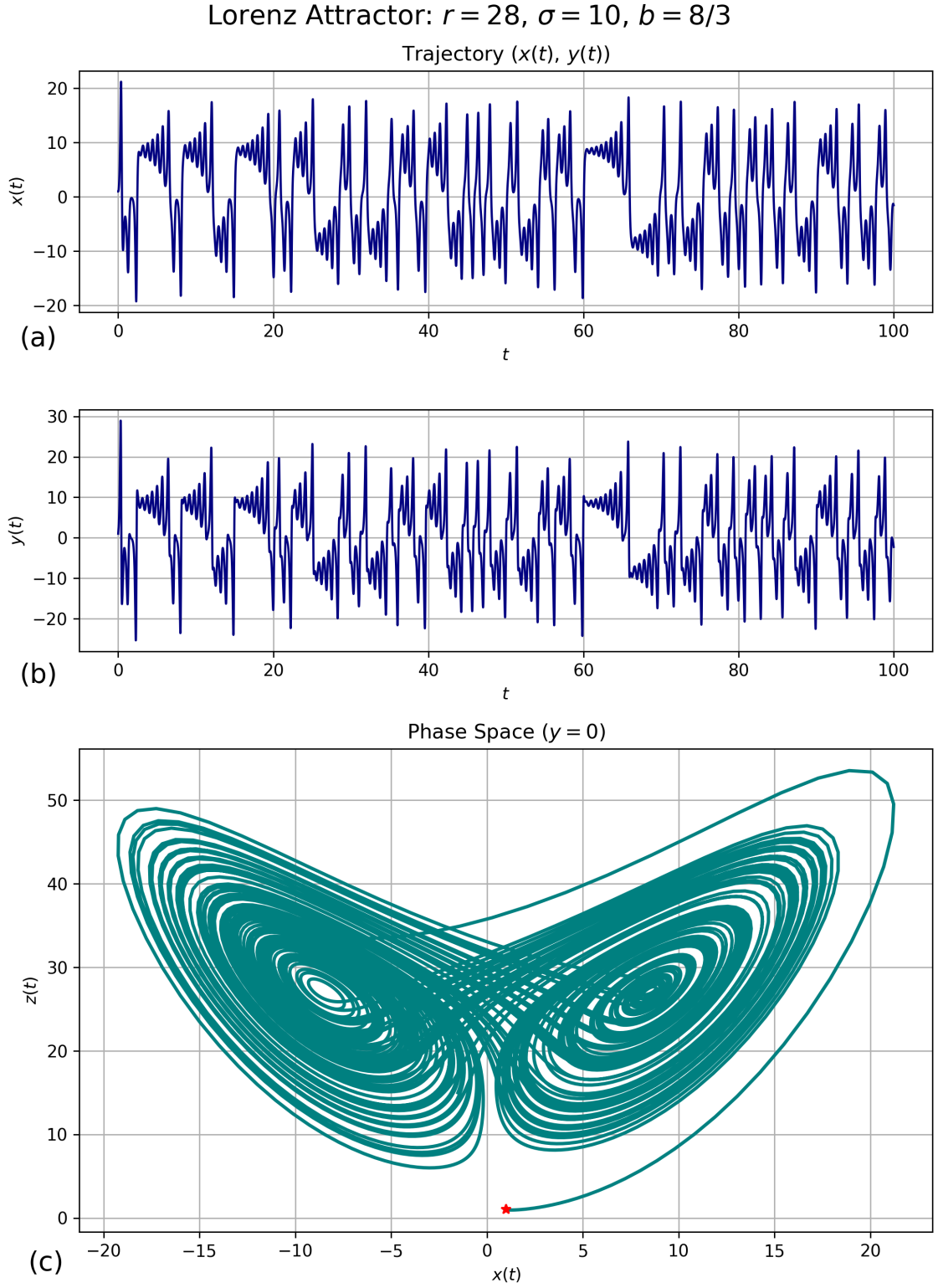


Figure 1: Lorenz Attractor system for parameters $r = 28$, $\sigma = 10$, and $b = 8/3$. Panel (a) displays the $x(t)$ coordinate for the trajectory, and panel (b) likewise depicts the $y(t)$ behavior. Panel (c) shows the trajectory of the dynamical system through system coordinates, projected into the $y = 0$ plane. The initial condition used to generate this trajectory is $\mathbf{x}_0 = (1, 1, 1)$, marked with the red star in panel (c). The timestep used for the integration procedure (see Section 2) is $\Delta t = 10^{-2}$.

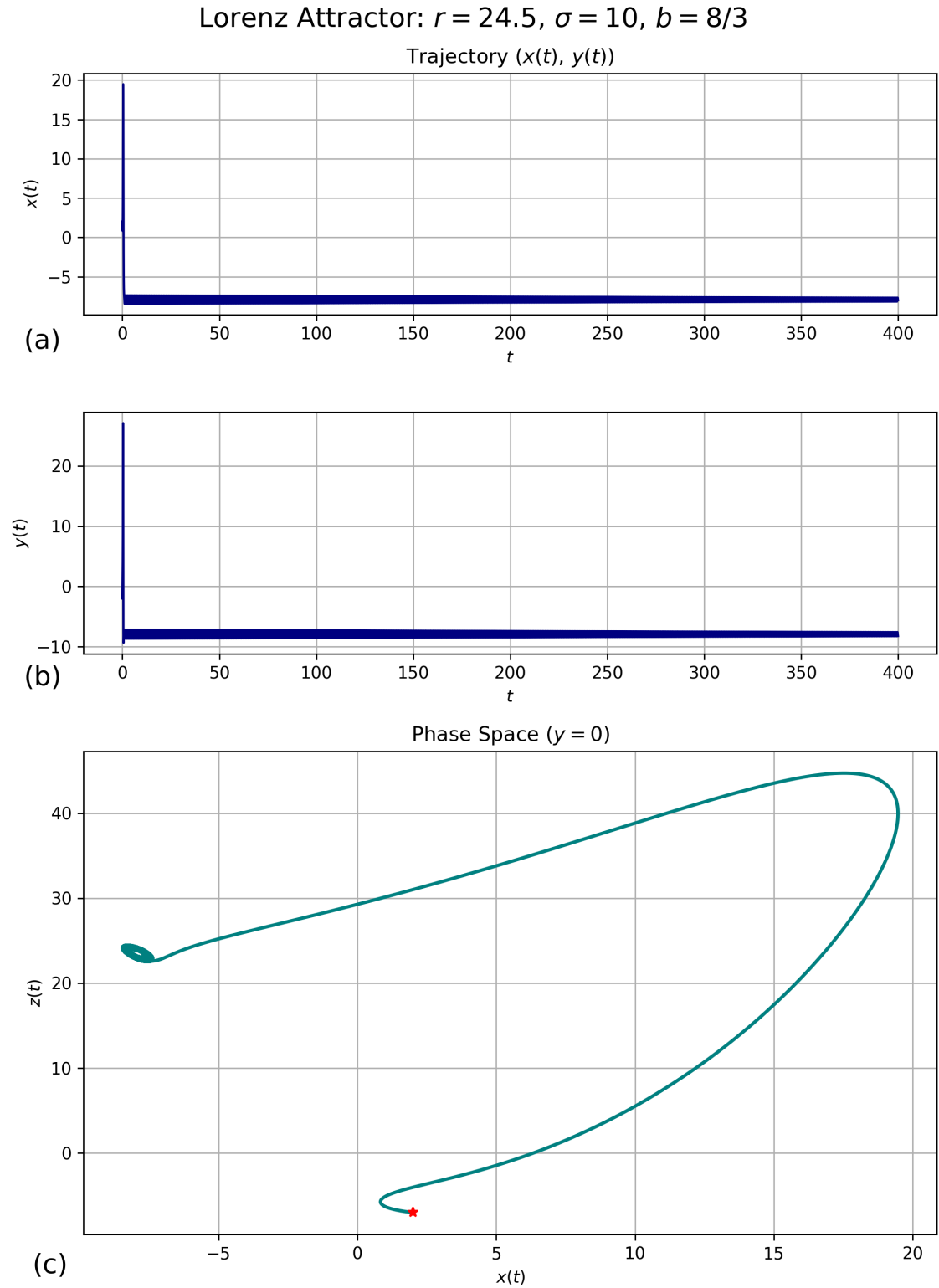


Figure 2: Lorenz Attractor system for parameters $r = 24.5$, $\sigma = 10$, and $b = 8/3$. Panel (a) displays the $x(t)$ coordinate for the trajectory, and panel (b) shows the $y(t)$ coordinate. Panel (c) shows the trajectory of the dynamical system through phase space coordinates, projected into the $y = 0$ plane. The initial condition used to generate this trajectory is $\mathbf{x}_0 = (2, -2, -7)$, marked with the red star in panel (c).

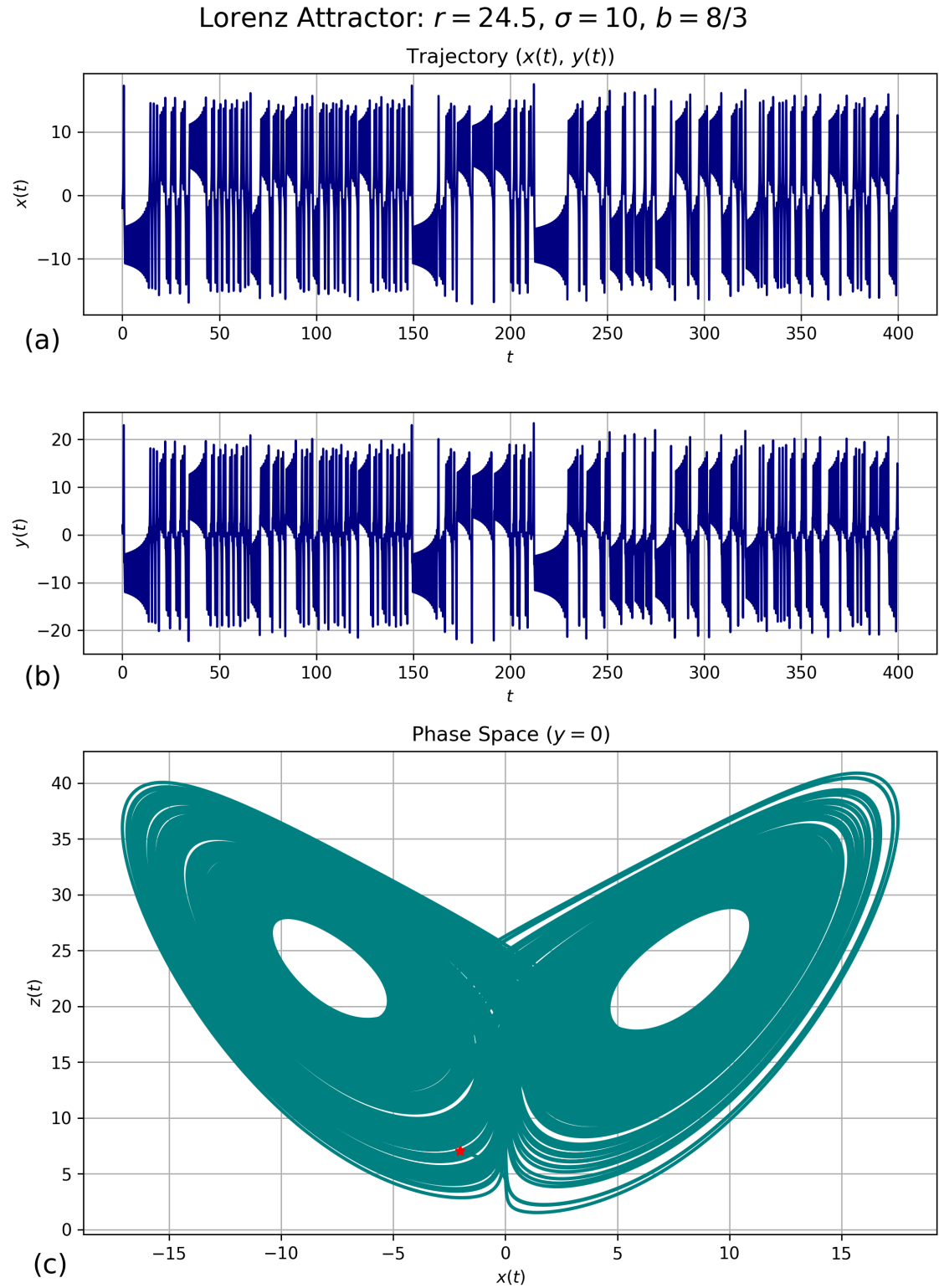


Figure 3: Lorenz Attractor system for parameters $r = 24.5$, $\sigma = 10$, and $b = 8/3$. Panel (a) displays the $x(t)$ coordinate for the trajectory, and panel (b) depicts the evolution of the $y(t)$ coordinate. Panel (c) shows the trajectory of the dynamical system through phase space coordinates, projected into the $y = 0$ plane. The initial condition used to generate this trajectory is $\mathbf{x}_0 = (-2, 2, 7)$, indicated by the red star in panel (c).

depicted in the $y = 0$ phase portrait (Figure 1(c)). This behavior is not strongly sensitive to the initial condition.

Now, we advance to the case assigned by Problem 2. We let $r \rightarrow 24.5$. It is critical to notice that this value of r is *slightly below* the critical value $r_h = 24.74$, at which point the stable fixed point turns into an unstable limit cycle via a supercritical Hopf bifurcation. The “typical” case shown in Figure 1 lives *above* this value with $r > r_h$: for that case, *all* trajectories that approach the fixed points will move back and forth between unstable limit cycles around each fixed point via the strange attractor. However, for values of $r < r_h$, there exist trajectories that can stably approach each of the pair of fixed points in finite time. Furthermore, the value of $r = 24.5$ is *slightly above* the critical value $r_c = 24.06$ at which the strange attractor possesses a finite timescale. That is, when r drops below $r_c = 24.06$, the system transitions from chaotic dynamics around a strange attractor to transient chaotic behavior, i.e., it displays meta-stable chaotic dynamics. Thus, our region of parameter space ($r = 24.5$) for the Lorenz system exhibits trajectories of both possibilities: either a stable orbit converging towards an attractive fixed point, or a chaotic trajectory that approaches the strange attractor and oscillates about both fixed points infinitely many times as $t \rightarrow \infty$.

In order to evaluate the behavior of the system for this set of parameters, we set out to identify different trajectories that exhibit these unique behaviors discussed above. Owing to the fundamental principles of chaos on strange attractors, we may identify and distinguish between trajectories that lie on and off of the strange attractor set by varying the initial conditions for the integration procedure. We first note that, analogous to the system when $r = 28$, any trajectories with an initial condition satisfying $\mathbf{x}_0 = (0, 0, \alpha)$ for an arbitrary $\alpha \in \mathbb{R}$ will “fall” monotonically along the stable manifold ($z = 0$) that corresponds to the fixed point at the origin until the fixed point is reached. Varying the initial configuration of the system to incorporate nonzero values of $x(0), y(0)$ allows us to recover trajectories that illustrate less trivial behavior.

An initial condition of $\mathbf{x}_0 = (2, -2, -7)$ was identified to converge towards an equilibrium, avoiding indefinite oscillations and exhibiting the stable behavior that becomes possible for $r < r_h$. This trajectory is shown in Figure 2, where the individual coordinate trajectories $x(t)$ and $y(t)$ (panels (a) and (b), respectively) illustrate two phases of the behavior. This is readily visible by inspection of the trajectory in panel (c): there is a transient phase and a steady-state phase. At first during the transient phase, the path of the trajectory veers “outward” in phase space along the stable manifold. In the steady-state phase, as it approaches the fixed point, it falls into a relaxed oscillation around the equilibrium. As indicated by the blue curves in Figures (a) and (b), the amplitude of the oscillations decreases with increasing time, until it reaches an approximately constant value in the immediate neighborhood of the equilibrium point. This occurs beyond the range displayed in Figure 2. Comparison of Figures 1 and 2 emphasize that these two trajectories encounter globally distinct fates.

On the other hand, an initial condition equal to $\mathbf{x}_0 = (-2, 2, 7)$ exhibits chaotic dynamics on the strange attractor as it evolves. This case is depicted in Figure 3, where the familiar behavior of the Lorenz system for $r = 28.5$ (from Figure 1) can be readily identified. In the entire 400 time units of evolution displayed in Figure 3, the trajectory shifts erratically between oscillations about either of the two equilibria dozens of times. This is consistent with the chaotic behavior on the strange attractor seen in Figure 1.

To summarize, by lowering the value of the r parameter below r_h while keeping it above r_c , this configuration of the Lorenz system is capable of admitting *both* trajectories which stably approach attracting equilibrium points as well as those that never settle to a steady-state but exhibit chaos on a strange attractor manifold.

3 Problem 9.3.8: Practice with the definition of an Attractor

Consider the following (familiar) system in polar coordinates:

$$\dot{r} = r(1 - r^2) \quad ; \quad \dot{\theta} = 1 \quad .$$

Let D be the disk $x^2 + y^2 \leq 1$.

3.1 Part (a): Is D an Invariant Set?

An invariant set in configuration space of a dynamical system is a set whose elements at time T remain within the set for all times $t > T$. Since $D := x^2 + y^2 = r^2 \leq 1$, trajectories inside the disc will all have values of $r < 1$. Such positions will be subject to $\dot{r} < 0$, and thus they will *remain* in the disc with $r < 1$ indefinitely. Hence, D is an invariant set of the system.

3.2 Part (b): Does D attract an open set of initial conditions?

For D to attract an open set of initial conditions, there must exist a set U such that $D \subset U$ and all trajectories initialized in U will tend toward D as $t \rightarrow \infty$.

Since an open set can be arbitrarily close in radius to the disc D so long as it is separated by a finite difference, we can check for values of $r > 1$ whether or not the time derivative in the radial coordinate bears towards D . For $r = 2$, for example, $\dot{r} = 2(1 - 4) = -6$: other values of $r > 1$ also correspond to $\dot{r} < 0$. Hence, trajectories initialized in a set U , of which D is a subset, are attracted towards D and approach it as $t \rightarrow \infty$. Ergo, D attracts an *open set* of initial conditions.

3.3 Part (c): Is D an Attractor? If not, why? If so, find its Basin of Attraction.

For D to be an attractor, it must satisfy the following criteria:

- D is an invariant set
- D attracts an open set of initial conditions
- D is minimal: there is no proper subset of D that satisfies the first two criteria

Condition #1 has been established in Section 3.1. Condition #2 has been established in Section 3.3. Thus, if D is minimal, then it meets all three conditions to be considered an attractor.

However, for D to be minimal, there must be no proper subset of D that satisfies the first two conditions. Since the ring $x^2 + y^2 = 1$ is a proper subset of D and it satisfies the above two criteria, D is not minimal. Therefore, D is *not* an attractor.

3.4 Part (d): Repeat part (c) for the circle $x^2 + y^2 = 1$.

Now we set out to establish the above three criteria for the unit circle $S := x^2 + y^2 = 1$. Since $\dot{r} = 0$ for $r = 1$, trajectories initialized on the unit circle will remain within the set S for all time. Hence, S is an invariant set. Analogous to the argument in section 3.2, the set S also attracts an open set of initial conditions, of which S is a proper subset. An example of the attracted open set could include other regions of the original set D . Finally, since the limit cycle at $r = 1$ represents all equilibrium behavior in the set S , this set is minimal. Hence, S is an open set.

To determine its basin of attraction, we consider the nature of the expression for $\dot{r}$. By inspection of a plot of $\dot{r}$ or the expression itself, one can show that r will increase as t increases for all $0 < r \leq 1$. For larger values, r will approach the $r = 1$ limit cycle from the “outside”. In any case, trajectories initialized anywhere but the origin will approach the limit cycle on S as time approaches infinity. Hence, the basin of attraction for the attractor S is the set

$$\mathbb{R} \times \mathbb{R} \setminus \{0, 0\} \quad .$$

4 Problem 9.3.9: Exponential Divergence

Using numerical integration of two nearby trajectories, estimate the largest Liapunov exponent for the Lorenz system, assuming that the parameters have their standard values: $r = 28$, $\sigma = 10$, and $b = 8/3$.

In order to calculate the largest Lyapunov exponent $\lambda_{\max}$ numerically, we turn to the procedure introduced in lecture. This involves the following steps:

1. Compute two trajectories of the system whose initial data are within the same neighborhood ($\delta(t=0) = \epsilon$)
2. Calculate the Euclidean distance $\delta(t)$ between the two trajectories as a function of time
3. Compute the logarithm of the separation $\log |\delta(t)|$ and determine the interval of time before the separation has reached a similar order to the size of the attractor manifold itself
4. Using a sufficiently high number of data points in this interval, fit the curve $(t, \log |\delta(t)|)$ to a linear polynomial and extract the slope
5. The extracted value is approximately the value of the largest Lyapunov exponent

We carry out this exact procedure using the integrator described in Section 2. To do so, we calculate the orbits for two initial positions: $(1, 1, 1)$ and $(1 + \epsilon, 1, 1)$, where $\epsilon = 10^{-10}$. Then, following the above steps, we extract a value of $\lambda_{\max} \approx 0.968$. The results for the trajectories, the separation distance curve $(t, \log |\delta(t)|)$, and the parameter extraction are depicted in Figure 4. As can be seen by panel (c), the exponential divergence of the trajectories occurs after an initially small difference starts to grow. This behavior persists until about $t = 38$, where the separation begins to level off. The roughly constant value the separation approaches corresponds loosely to the “size” of the attractor manifold in phase space. Hence, we restrict the time interval for the fit and parameter extraction to the portion that exhibits the clear exponentially-divergent behavior, shaded blue in Figure 4(c). The value of $\lambda_{\max} \approx 0.968$ is consistent with reported values in Strogatz and in the literature.

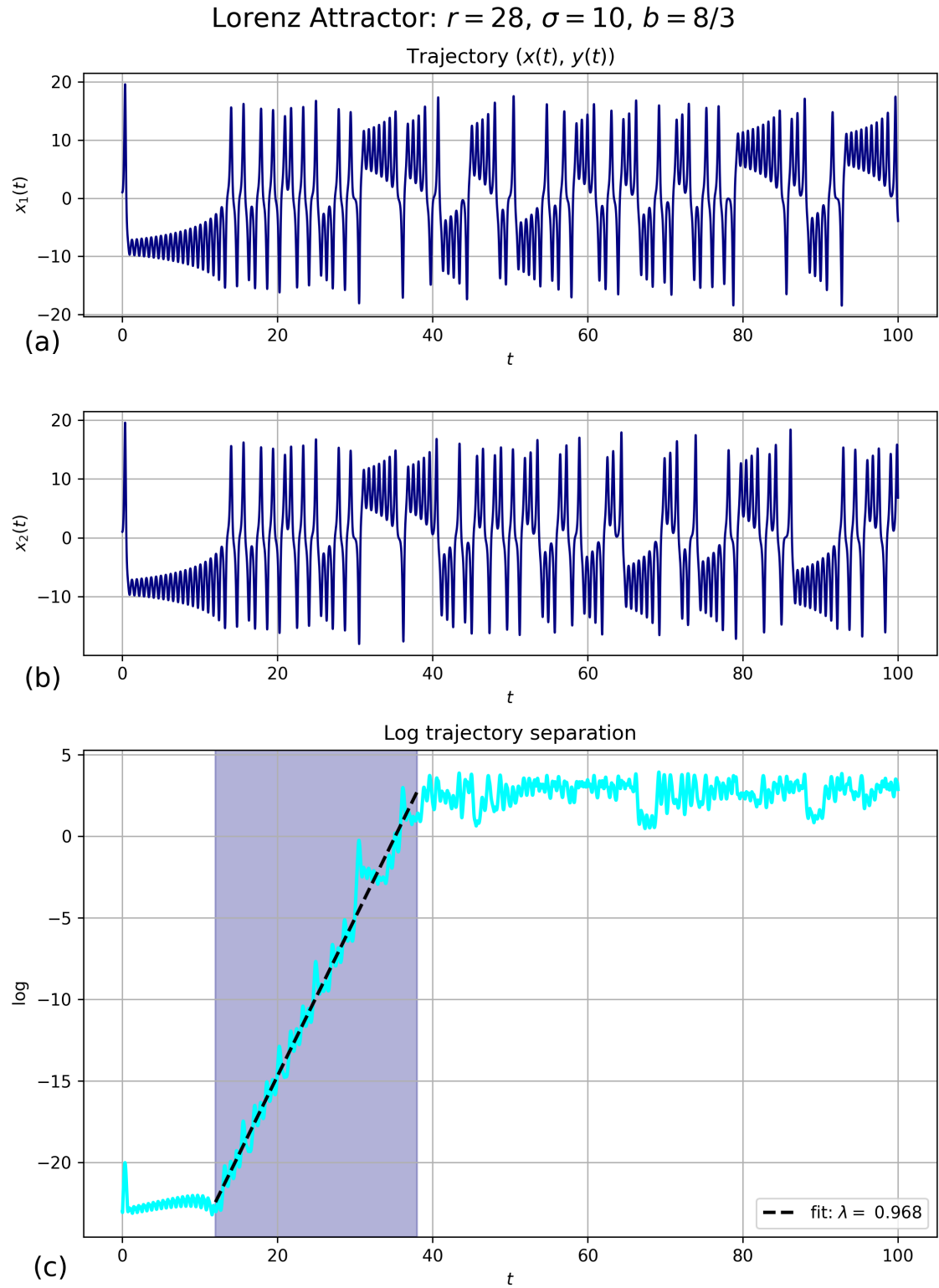


Figure 4: Extraction of the maximal Lyapunov exponent for the Lorenz system with parameters $r = 28$, $\sigma = 10$, and $b = 8/3$. Panel (a) displays the $x_1(t)$ coordinate for the “original” trajectory, and panel (b) shows the analogous evolution $x_2(t)$ for the perturbed trajectory. Panel (c) shows the logarithm of the trajectory separation. The blue shaded region represents the time interval used to extract the fit, and the fit is included as the black dashed line.

5 Software Usage

The entirety of the work was carried out on pen and paper prior to typesetting, except for the integration of the Lorenz system (see Section 2), as well as plotting: these were completed using python (`numpy` and `matplotlib`) libraries.